



OBJECT-ORIENTED PROGRAMMING (CS 213)

DR. DAVID EBO ADJEPONG-YAMOA

PROJECT DESIGN DRAFT MEMBERS:

**KELVIN PAA EKOW WILSON , RONALD SELASIE OCLOO,
DARRELL YRAM FIAM-COBLAVIE , JADZIA AFIA OHENEWAA MANTEY**

Project title: Player Performance Ranking System

DESCRIPTION

The 2026 FIFA World Cup is one of the most anticipated events this year featuring 48 different countries across all continents with the hosts being the United States, Mexico and Canada. The World Cup is one of the sporting events which attracts a lot of people and improves socialization for fans and supporters. Player ranking is one of the most important aspects of this high-staged event. Supporters and fans gamble on which player comes out with the most goals and assists, the overall best player, the keeper with the most saves and clean sheets and the FIFA board rewards all players who emerge victorious in this category.

Most times, challenges are faced when officials want to manually track player performances and rank the players based on their data, which sometimes causes inaccurate calculations. This is the problem we are trying to fix and this is where our system comes in handy.

Our system allows officials to add all the world cup registered players to our database by providing the following information about each player.

- Player name
- Player Country
- Player position
- Player club
- Player FIFA ID

After every match, officials can come and input their match statistics about specific players according to the metrics of goals, assists, yellow cards, red cards, tackles, passes completed, tackles, clean sheets and saves.

At the end of the tournament, officials get a ranked list of players with the most points in the respective metrics.

To find the overall best player at the end of the entire tournament, the weighted sum technique is applied. The player with the most goals or the most assists is definitely not the overall best player of the tournament so to find the overall best, a weight is assigned to each metric and each player's points are multiplied by the weights of each metric based on its importance and the total sum of all for each player is calculated. After which the data for each player will be grouped and ranked from the highest to the lowest. The weighted sum technique was chosen because it is very clear and easy to understand, its implementation is easy which makes the weights against some metrics easily adjustable.

PROPOSED SOLUTION FUNCTIONALITY

Our system is a user-friendly system which interacts with the user in the form of a menu where the user chooses whether to add a player, update the statistics, view rankings and exit the menu.

Overview of User interaction with the menu and detailed algorithm.

ADD PLAYER:

The user has to provide the following details.

- Player name
- Current club
- Position
- Nationality - Fifa id

For each detail, the user input is checked and validated to see if it matches the expected input. If it matches, an object is created and stored with the data entered by the user, and if it does not, the user is asked to re-enter the data. After creating the object, it is added to an arraylist and also written into a file.

UPDATE THE STATISTICS:

The user must choose which metric to update.

- Goals
- Assists
- Saves
- Clean sheets
- Yellow cards
- Red cards
- Tackles
- Passes completed

For each metric, the user must enter the name and FIFA ID of the player. The system checks if the player exists. If the player exists, the user proceeds to enter the number to be updated with and input validation is checked. Otherwise, the system redirects the user to the main menu to add the player.

RANKINGS:

- Goal Ranking
- Assist Ranking
- Passes completed Ranking
- Clean sheet Ranking
- Yellow Card Ranking
- Red Card Ranking
- Saves Ranking
- Tackles ranking

- Overall player Rankings

The Ranking option in the menu provides the user with various options to view the various rankings. Each ranking metric contains a sorted list of the players in descending order from who has the most and who has the least attribute. For the overall player ranking, the weighted sum technique is applied, where the players are sorted and ranked in descending order from who has the most points to the least points based on a certain criteria and after, the ranking is written into a file.

EXIT

This option on the menu allows the user to terminate the program. A goodbye message is displayed and the program ends.

CLASS DESCRIPTION

Player class is the super class which holds information about the player. It contains the following information.

- Player name
- Player Country
- Player position
- Player club
- Player FIFA ID
- Goals
- Assists

- Saves
- Cleansheets
- Redcards
- Yellowcards
- PassessCompleted
- Tackles.

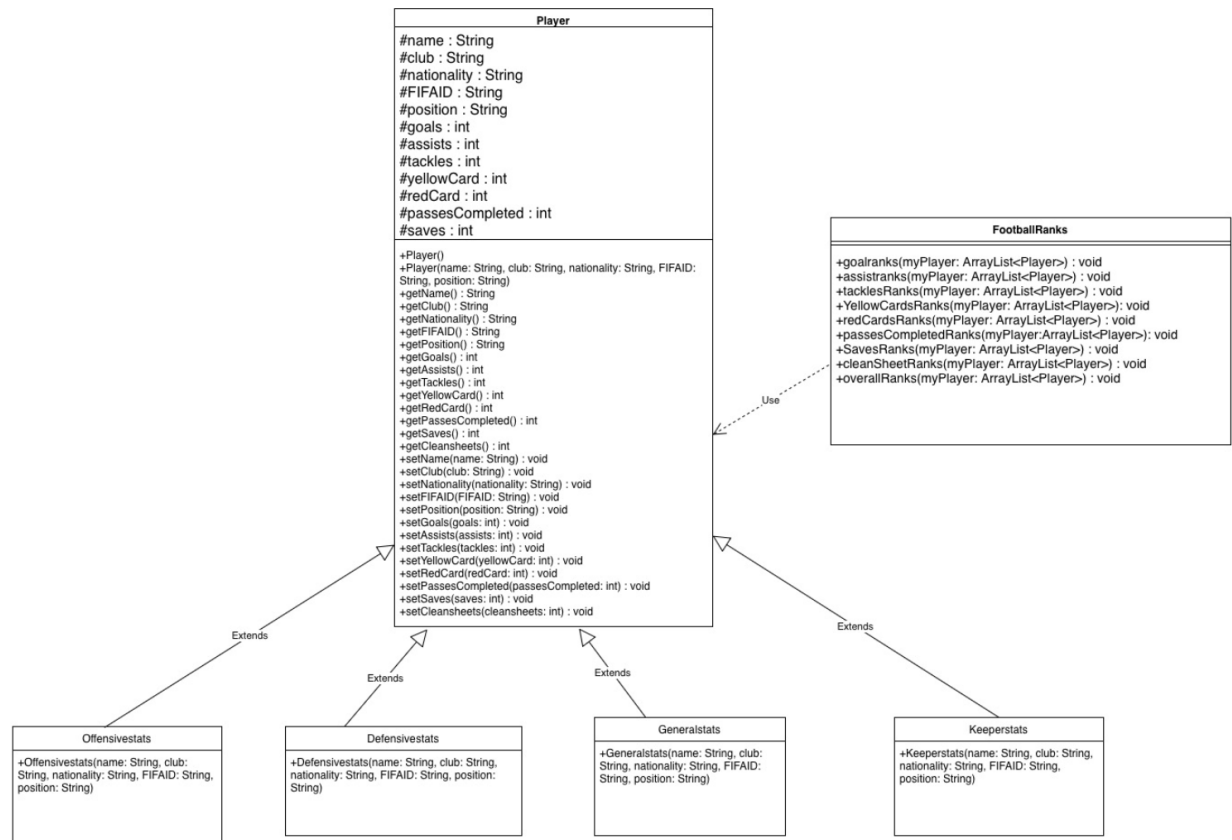
SUB CLASSES OF THE SUPER CLASS (PLAYER)

- Generalstats class
- Offensivestats class
- Keeperstatsclass
- Defensivestats class

The FootballRanks is the class which contains all the methods used for ranking the players according to the various metrics as well as overall best player ranking in which the weighted sum technique is applied to find the overall best player. All rankings are written into a file to be accessed at any time.

The main class contains the menu and manages all user inputs.

UML DIAGRAM



References

GeekforGeeks. Weighted Sum Method - Multicriteria Decision Making.

<https://www.geeksforgeeks.org/dsa/weighted-sum-method-multi-criteria-decision-making/>